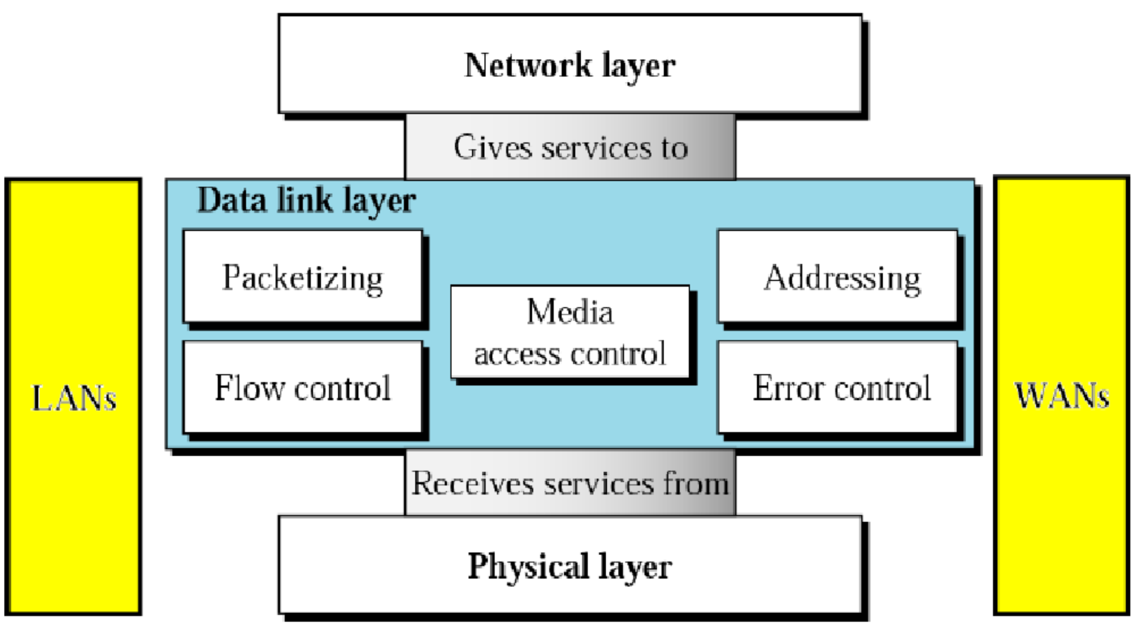
3.1 Data link layer functions - Framing, error control, flow control, access control

For reliable communication between two directly connected stations, the data link layer performs several key tasks:

- **Frame Synchronization:** Data is transmitted in blocks called frames. The start and end of each frame must be clearly marked so the receiver can distinguish them.
- **Flow Control:** Ensures the sender does not transmit frames faster than the receiver can process.
- **Error Control:** Detects and corrects bit errors introduced during transmission.
- **Addressing:** Identifies the sender and receiver stations involved in communication.
- **Control and Data Sharing:** Allows both control information and user data to travel over the same link.
- **Link Management:** Handles the setup, maintenance, and termination of a communication session, requiring coordination between stations.



Framing

Framing separates a message from source to a destination by adding a sender address and destination address. Message is divided into smaller frames so that a single-bit error affects only that small frame. Frames can be of fixed size or variable size.

- **Fixed-Size Framing**: The frame is of fixed size and there is no need to provide boundaries to the frame, length of the frame itself acts as delimiter. It suffers from internal fragmentation if data size is less than frame size.

- **Variable size framing:** In this there is need to define end of frame as well as beginning of next frame to distinguish. This can be done in two ways:

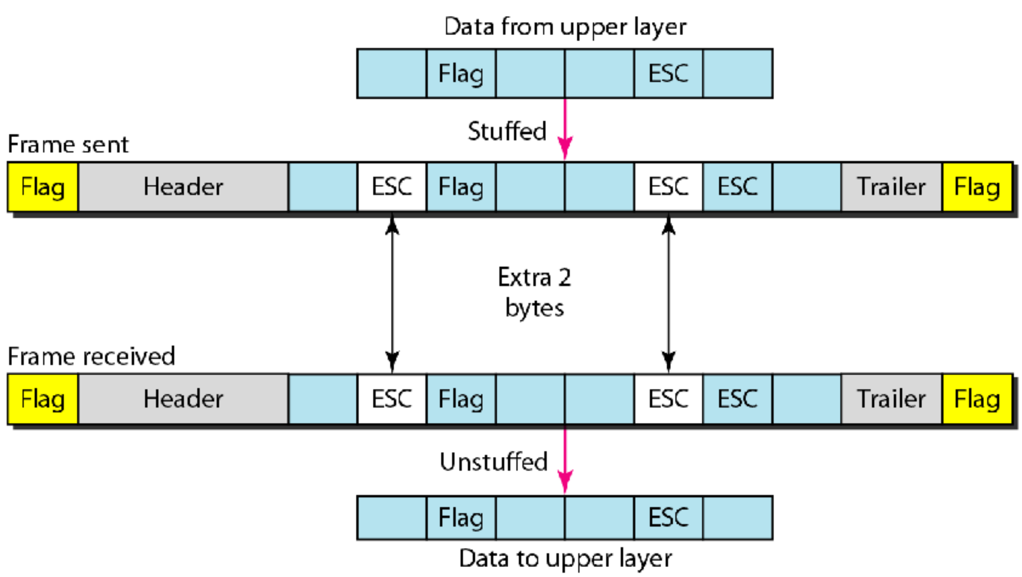
  - **Length field** – We can introduce a length field in the frame to indicate the length of the frame. Used in **Ethernet (802.3)**. The problem with this is that sometimes the length field might get corrupted.

  - **End Delimiter (ED)** – We can introduce an ED (pattern) to indicate the end of the frame. Used in **Token Ring**. The problem with this is that ED can occur in the data. This can be solved by Character byte stuffing and bit stuffing

CHARACTER/BYTE STUFFING:

- Byte stuffing is the process of adding 1 extra byte whenever there is a flag or escape character in the text.

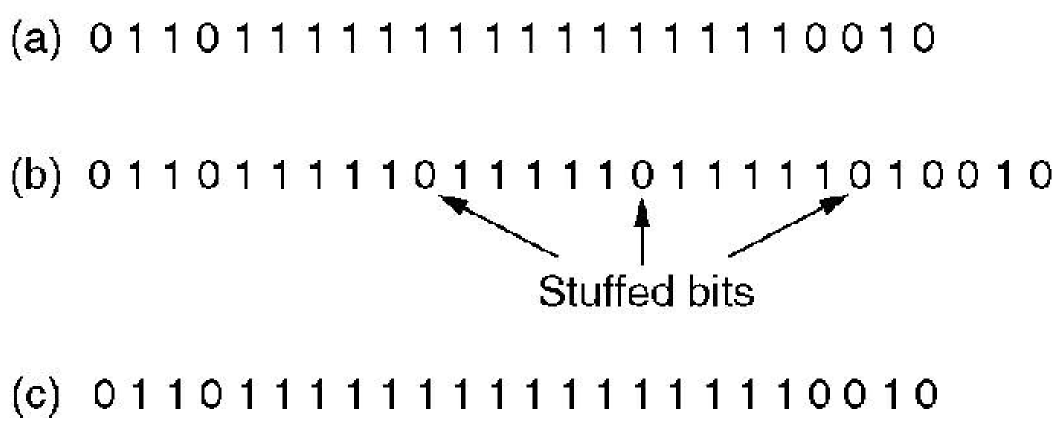
- **Disadvantage –** It is very costly and obsolete method.



BIT STUFFING

Let **ED = 11110** and suppose the **data = 11110**

- To prevent confusion with the end delimiter, the sender inserts a 0 inside the pattern.
- So the transmitted data becomes: **111100**
- The receiver gets the frame.
- When the receiver detects the pattern **111100**, it understands that the extra 0 was inserted intentionally.
- The receiver removes the stuffed 0 and restores the original data: **11110**



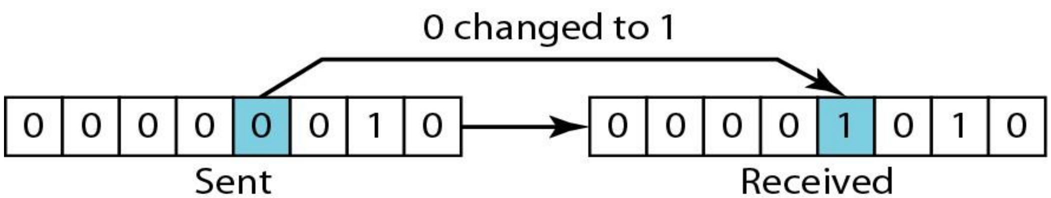
3.2 Error control codes: Parity, checksum, CRC

- Ensure error free transmission in spite of error probability in the physical layer by adding error control bits as trailer
- Data can be corrupted during transmission (because of any reason)
- Error: Whenever bits flow from one point to another, they are subject to unpredictable changes because of interference; this interference can change the shape of the signal -> change of corresponding bit(s)
- Some mechanism is required for detection and correction of such possible errors

Types of Errors

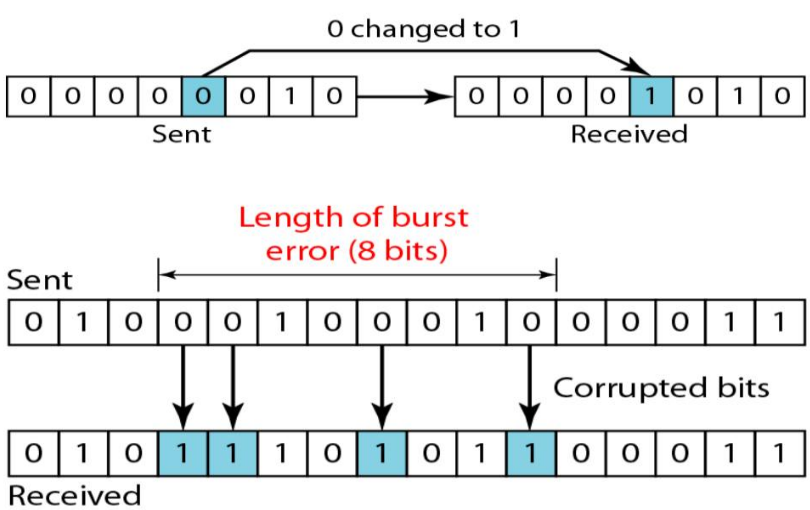
Single-bit Error: Only one bit in the data unit has changed

- In a single-bit error, only **one bit** in a given data unit (such as a byte, character, or packet) is changed from 1 to 0 or from 0 to 1.
- **Cause:** Usually caused by white noise.
- **Example:** Sent 00110110 → Received 001101**0**0

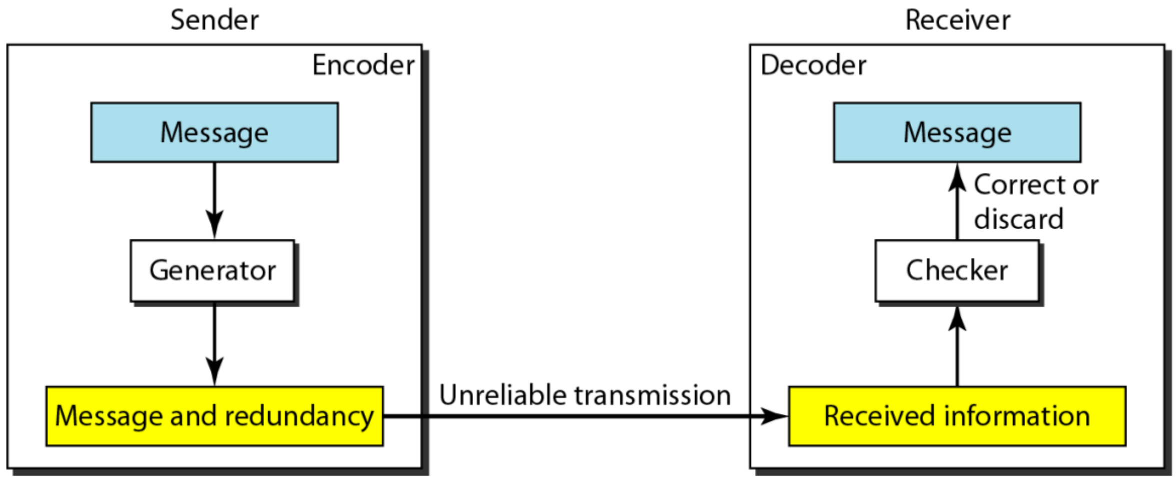


Burst Error: Two or more bits in the data unit have changed

- In a burst error, **two or more bits** in the data unit have changed. It doesn't mean consecutive bits are corrupted; it means the error spans from the first corrupted bit to the last corrupted bit.
- **Cause:** Usually caused by impulse noise (lightning, sudden electrical spikes) or fading.
- **Example:** Sent 00110110 → Received 0**1**1**01**1**0**0 (Length of burst is 5).



➢ To detect or correct errors, we need to send extra (redundant) bits with data



Parity-bit

A **parity bit** is a simple error-detection method used in data communication. It adds **one extra bit** to a group of bits (usually a byte) to help detect transmission errors.

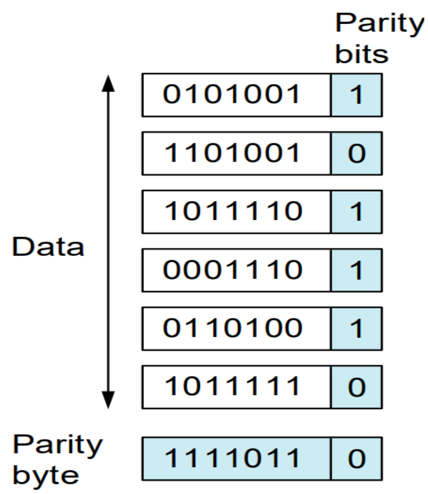
Types of Parity

1. **Even Parity**
   The total number of 1s (including the parity bit) must be **even**.

Example:
Data = 1011001
Number of 1s = 4 (already even)
So parity bit = 0
Transmitted data = 10110010
If one bit changes during transmission, the number of 1s becomes odd → error detected.

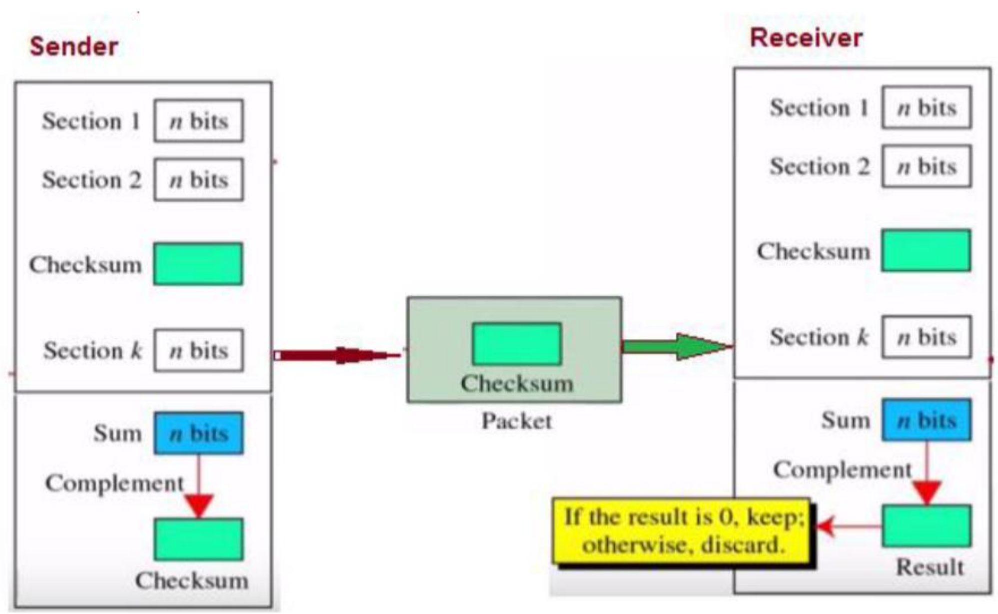
2. **Odd Parity**
   The total number of 1s (including the parity bit) must be **odd**.

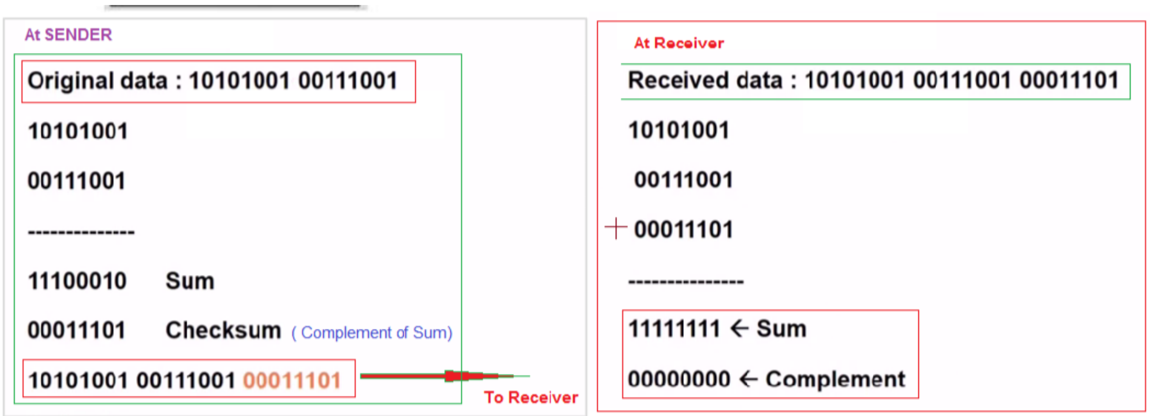
Example:
Data = 1011001
Number of 1s = 4 (even)
To make it odd, parity bit = 1
Transmitted data = 10110011



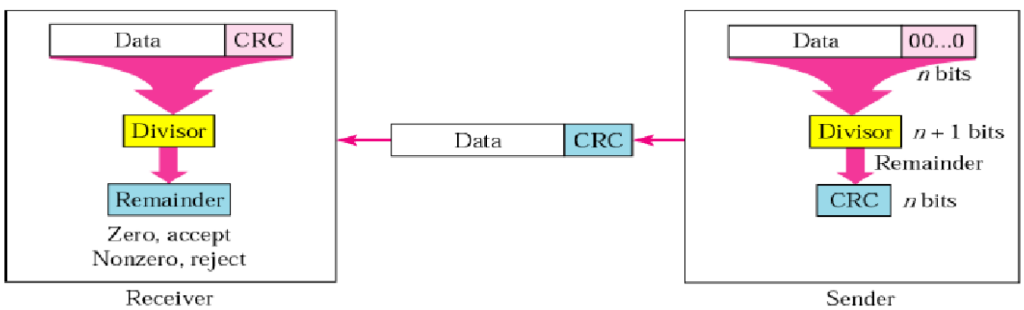
Checksum

The checksum is used in the Internet by several protocols although not at the data link layer.
To create checksum, the sender does the following :-
- The unit is divided into K sections, each of n bits
- Section 1 and 2 are added together using 1's complement.
- Section 3 is added to the result of the previous step.
- Section 4 is added to the result of the previous step.
- The process repeats until section k is added to the result of previous step.
- The final result is complemented to make checksum



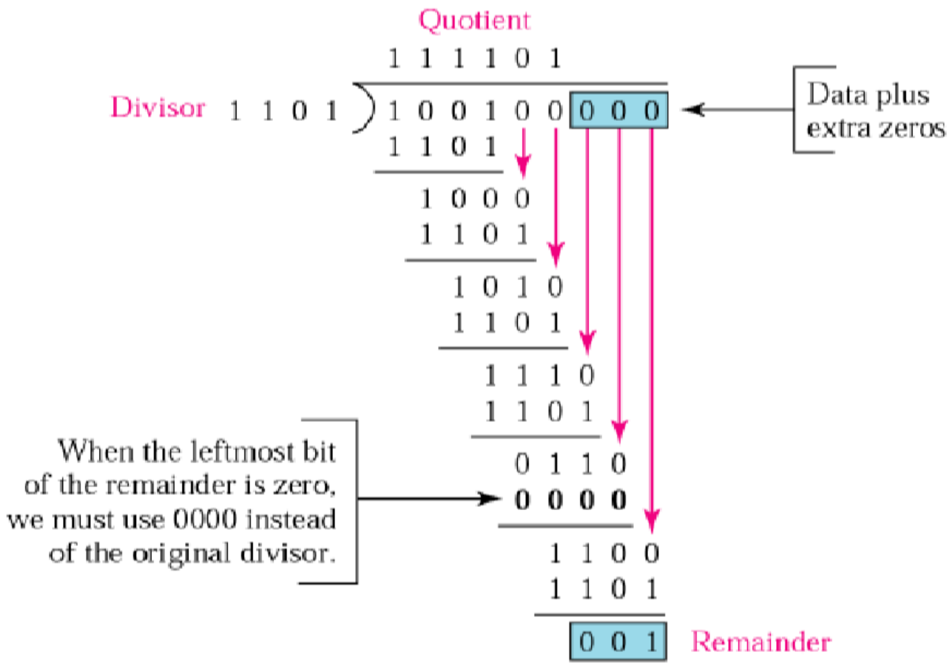


CRC(Cyclic Redundancy Check)

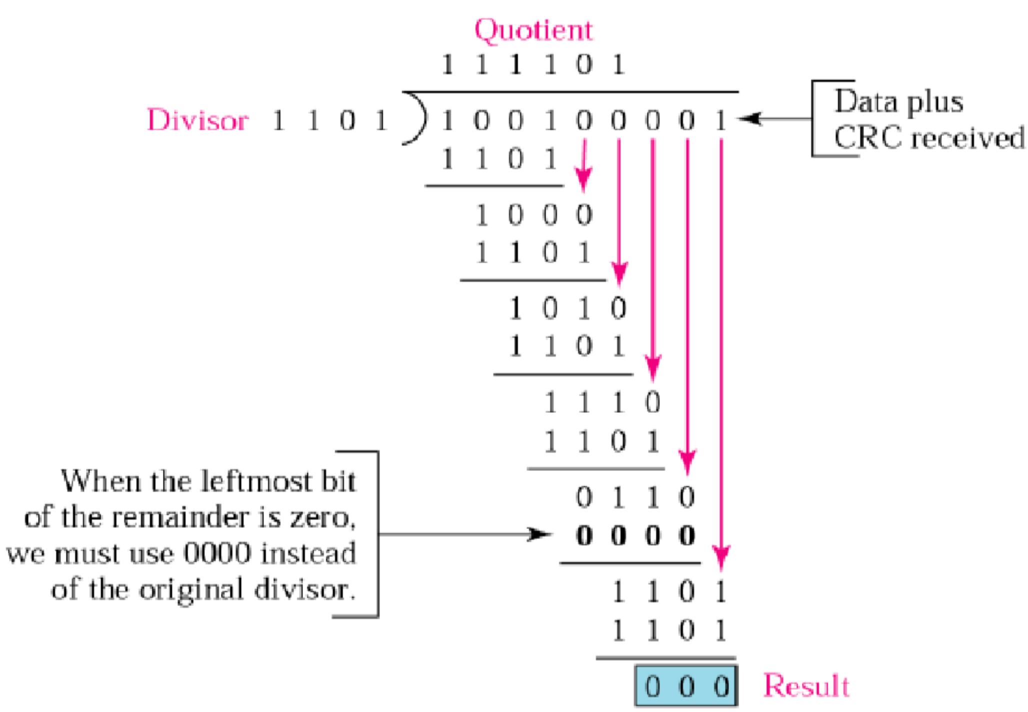


- The data or polynomial is appended with number of zeros equal to the degree of generator (divisor).
- The polynomial formed is divided by the divisor.
- The remainder of the division will be the value of CRC that will replace the data plus extra zeros i.e., remainder is added to the appended polynomial .
- This value is now transmitted to the receiver as the transmitted frame.
- At the receiver side, the data string and the CRC value is divided by the same value of divisor in the sender part.
- Then the remainder determines either to accept or reject the received data bit string.
- If the remainder is zero, then the data will be accepted or else it will be rejected.

CRC Encoding



CRC Decoding



Error Correcting Code

- Error-correcting codes are widely used on wireless links, which are notoriously noisy and error prone when compared to copper wire or optical fibers.
- Without error-correcting codes, it would be hard to get anything through.
- However, over copper wire or fiber, the error rate is much lower, so error detection and retransmission are usually more efficient there for dealing with the occasional error.
- Hamming Code is Example of error Correcting Code

Hamming Code

Dec no.	Binary	P1	P2	P4	P8
1	0001	X			
2	0010		X		
3	0011	X	X		
4	0100			X	
5	0101	X		X	
6	0110		X	X	
7	0111	X	X	X	
8	1000				X
9	1001	X			X
10	1010		X		X
11	1011	X	X		X
12	1100			X	X
13	1101	X		X	X
14	1110		X	X	X
15	1111	X	X	X	X

P1= (1,3,5,7,9,11,13,15) -> first least significant bit

P2= (2,3,6,7,10,11,14,15) -> second least significant bit

P4= (4,5,6,7,12,13,14,15) -> Third least significant bit

P8= (8,9,10,11,12,13,14,15) -> Fourth least significant bit

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
		1		2	3	4		5	6	7	8	9	10	11

Pos	$2^0=1$	$2^1=2$	$2^2=4$	$2^3=8$
1	1	0	0	0
2	0	1	0	0
3	1	1	0	0
4	0	0	1	0
5	1	0	1	0
6	0	1	1	0
7	1	1	1	0
8	0	0	0	1
9	1	0	0	1
10	0	1	0	1
11	1	1	0	1
12	0	0	1	1
13	1	0	1	1
14	0	1	1	1
15	1	1	1	1

Data To be encoded 10101101011

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
		1		0	1	0		1	1	0	1	0	1	1

- After placing the data in the table we find that in positions 3, 6, 9, 10, 12, 14 and 15 we have a ‘1’. Using our previous conversion table we obtain the binary representation for each of these values.
- We then exclusive OR the resulting values (essentially setting the parity bit to 1 if an odd # of 1’s else setting it to 0). Hamming Code 20
- After placing the data in the table we find that in positions 3, 6, 9, 10, 12, 14 and 15 we have a ‘1’. Using our previous conversion table we obtain the binary representation for each of these values.
- We then exclusive OR the resulting values (essentially setting the parity bit to 1 if an odd # of 1’s else setting it to 0).

	1	1	0	0	3
	0	1	1	0	6
	1	0	0	1	9
	0	1	0	1	10
	0	0	1	1	12
	0	1	1	1	14
	1	1	1	1	15
XOR	1	1	0	1	(11)

- Encoded Data

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	1	0	0	1	0	1	1	1	0	1	0	1	1

1	2	3	4	5	6	7	8	9	10	11	12	13	14	15
1	1	1	0	0	1	0	1	1	1	0	1	0	0	1

	1	1	0	0	3
	0	1	1	0	6
	1	0	0	1	9
	0	1	0	1	10
	0	0	1	1	12
	1	1	1	1	15
XOR	1	0	1	0	

	1	1	0	1	sent/received
	1	0	1	0	new calculated
XOR	0	1	1	1	this bit was flipped (14)

Error Detection Process

- **Transmitter**
  - For a given frame, an error-detecting code (check bits) is calculated from data bits
  - Check bits are appended to data bits

- **Receiver**
  - Separates incoming frame into data bits and check bits
  - Calculates check bits from received data bits
  - Compares calculated check bits against received check bits
  - Detected error occurs if mismatch

Flow Control

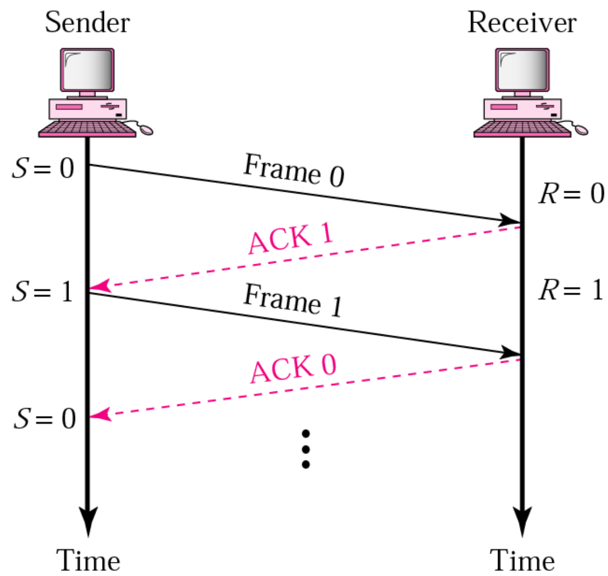
- Flow control coordinates the amount of data that can be sent before receiving acknowledgement
- It is one of the most important functions of data link layer.
- Flow control is a set of procedures that tells the sender how much data it can transmit before it must wait for an acknowledgement from the receiver.
- Receiver has a limited speed at which it can process incoming data and a limited amount of memory in which to store incoming data.
- Receiver must inform the sender before the limits are reached and request that the transmitter to send fewer frames or stop temporarily.
- Since the rate of processing is often slower than the rate of transmission, receiver has a block of memory (buffer) for storing incoming data until they are processed.

ERROR AND FLOW CONTROL MECHANISMS

- Stop-and-Wait
- Sliding Window
- Go-Back-N ARQ
- Selective-Repeat ARQ

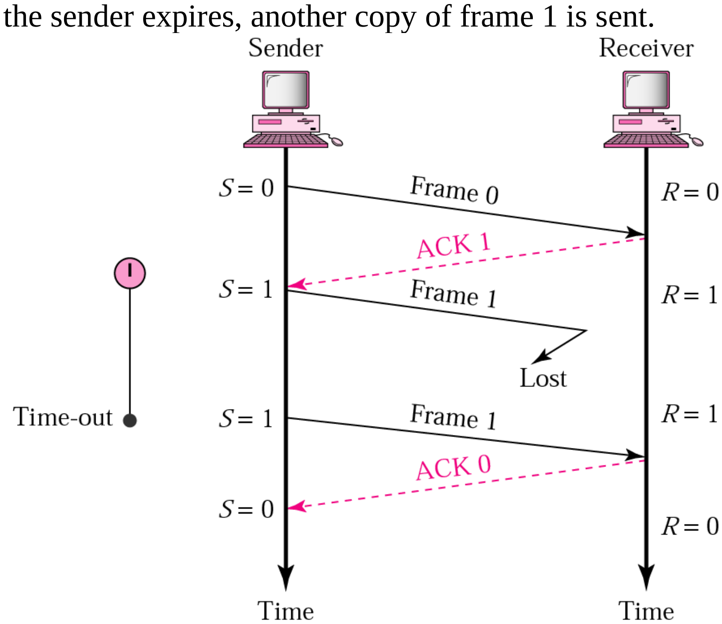
I. **STOP-AND-WAIT FLOW Control**

- A source transmits a frame.
- After the destination entity receives the frame, it indicates its willingness to accept another frame by sending back an acknowledgement to the frame just received..
- The source must wait until it receives the acknowledgement before sending the next frame.
- Receiver has a control variable that holds the number of the next frame expected (0 or 1).
- The process of alternately sending and waiting repeats until the sender transmits the end of transmission(EoT) frame.
- The destination can thus stop the flow of data simply by withholding acknowledgement..
- This procedure is inherently a half-duplex because the transmission is in both direction but only in one direction at a time.

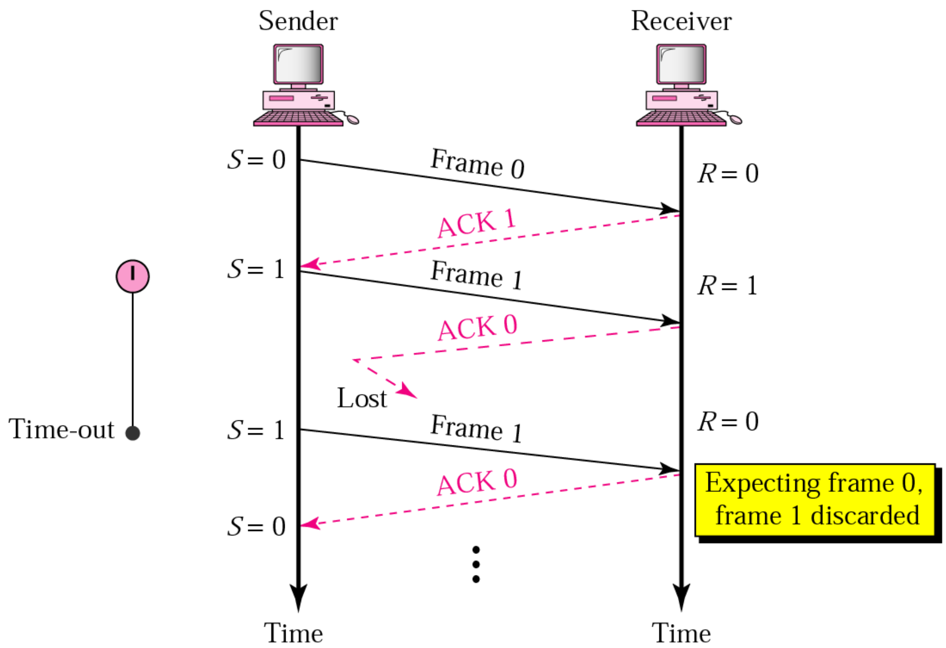


II. STOP-AND-WAIT ARQ :LOST ACK FRAME

- When a receiver receives a damaged frame, it discards it and keeps its value of R.
- After the timer at the sender expires, another copy of frame 1 is sent.



- If the sender receives a damaged ACK, it discards it.
- When the timer of the sender expires, the sender retransmits frame 1.
- Receiver has already received frame 1 and expecting to receive frame 0 (R=0). Therefore it discards the second copy of frame 1.



III. STOP-AND-WAIT ARQ: DELAYED ACK FRAME

- The ACK can be delayed at the receiver or due to some problem
- It is received after the timer for frame 0 has expired.
- Sender retransmitted a copy of frame 0. However, R =1 means receiver expects to see frame 1. Receiver discards the duplicate frame 0.
- Sender receives 2 ACKs, it discards the second ACK.

DISADVANTAGE OF STOP-AND-WAIT

- In stop-and-wait, at any point in time, there is only one frame that is sent and waiting to be acknowledged.
- This is Slow mechanism.
- To improve efficiency, multiple frames should be in transition while waiting for ACK

IV. GO-BACK-N ARQ

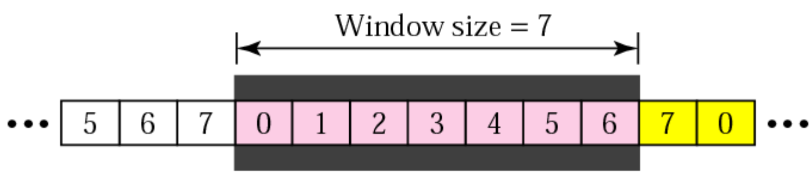
- We can send up to W frames before worrying about ACKs.
- We keep a copy of these frames until the ACKs arrive.
- This procedure requires additional features to be added to Stop-and-Wait ARQ.

V. SEQUENCE NUMBERS

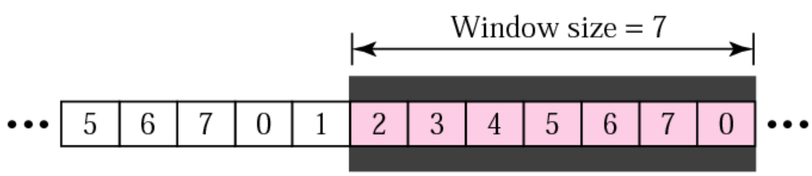
- Frames from a sender are numbered sequentially.
- We need to set a limit since we need to include the sequence number of each frame in the header.
- If the header of the frame allows m bits for sequence number, the sequence numbers range from 0 to $2^m - 1$. for m = 3, sequence numbers are: 1, 2, 3, 4, 5, 6, 7.
- We can repeat the sequence number.
- Sequence numbers are:
- 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, 2, 3, 4, 5, 6, 7, 0, 1, …

VI. SENDER SLIDING WINDOW

- At the sending site, to hold the outstanding frames until they are acknowledged, we use the concept of a window.
- The size of the window is at most 2^m -1 where m is the number of bits for the sequence number.
- Size of the window can be variable, e.g. TCP.
- The window slides to include new unsent frames when the correct ACKs are received



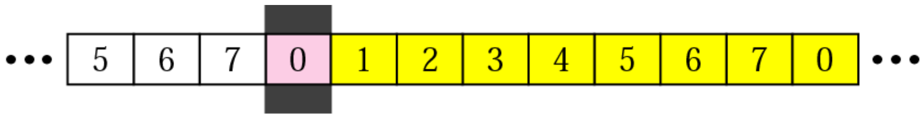
a. Before sliding



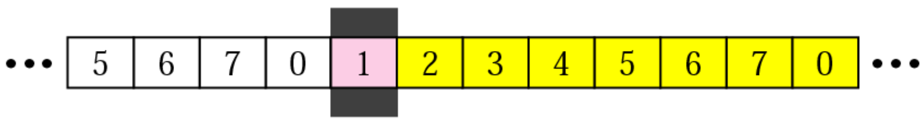
b. After sliding two frames

VII. RECEIVER SLIDING WINDOW

- Size of the window at the receiving site is always 1 in this protocol.
- Receiver is always looking for a specific frame to arrive in a specific order.
- Any frame arriving out of order is discarded and needs to be resent.
- Receiver window slides as shown in fig. Receiver is waiting for frame 0 in part a.



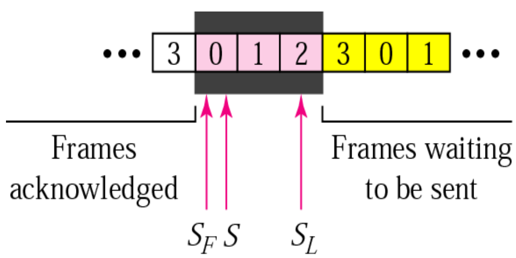
a. Before sliding



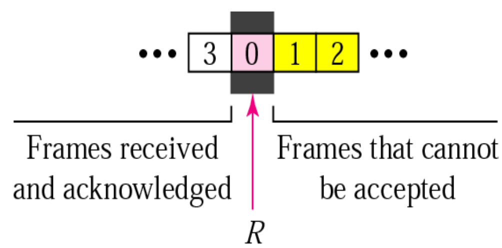
b. After sliding

CONTROL VARIABLES

- Sender has 3 variables: S, S_F, and S_L
- S holds the sequence number of recently sent frame
- S_F holds the sequence number of the first frame
- S_L holds the sequence number of the last frame
- Receiver only has the one variable, R, that holds the sequence number of the frame it expects to receive. If the seq. no. is the same as the value of R, the frame is accepted, otherwise rejected.



a. Sender window



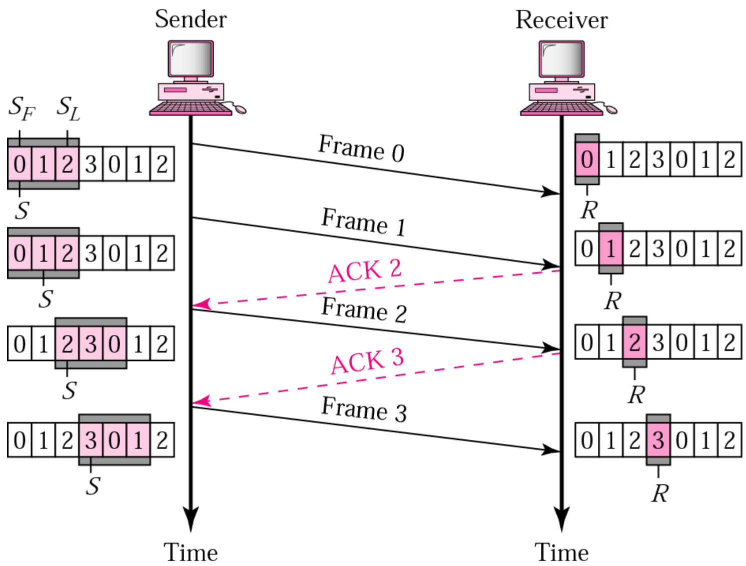
b. Receiver window

ACKNOWLEDGEMENT

- Receiver sends positive ACK if a frame arrived safe and in order.
- If the frames are damaged/out of order, receiver is silent and discard all subsequent frames until it receives the one it is expecting.
- The silence of the receiver causes the timer of the unacknowledged frame to expire.
- Then the sender resends all frames, beginning with the one with the expired timer.
- For example, suppose the sender has sent frame 6, but the timer for frame 3 expires (i.e. frame 3 has not been acknowledged), then the sender goes back and sends frames 3, 4, 5, 6 again. Thus it is called Go-Back-N-ARQ
- The receiver does not have to acknowledge each frame received, it can send one cumulative ACK for several frames.

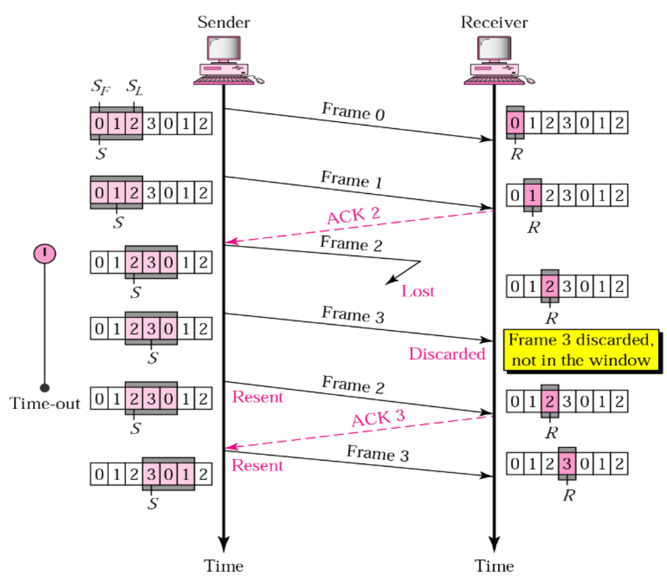
GO-BACK-N ARQ, NORMAL OPERATION

- The sender keeps track of the outstanding frames and updates the variables and windows as the ACKs arrive.



GO-BACK-N ARQ, LOST FRAME

- Frame 2 is lost
- When the receiver receives frame 3, it discards frame 3 as it is expecting frame 2 (according to window).
- After the timer for frame 2 expires at the sender site, the sender sends frame 2 and 3. (go back to 2)



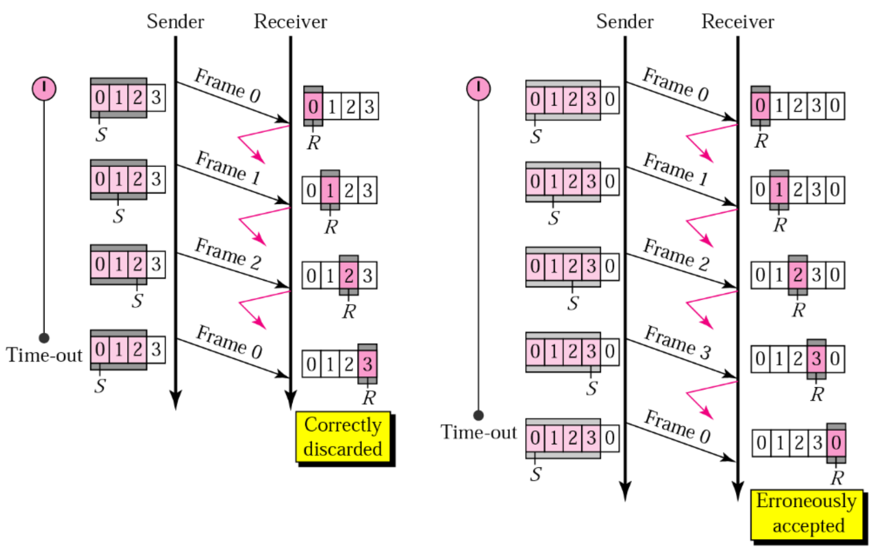
GO-BACK-N ARQ, DAMAGED/LOST/DELAYED ACK

- If an ACK is damaged/lost, we can have two situations:
- If the next ACK arrives before the expiration of any timer, there is no need for retransmission of frames because ACKs are cumulative in this protocol.

- If ACK1, ACK2, and ACk3 are lost, ACK4 covers them if it arrives before the timer expires.
- If ACK4 arrives after time-out, the last frame and all the frames after that are resent.
- Receiver never resends an ACK.
- A delayed ACK also triggers the resending of frames

GO-BACK-N ARQ, SENDER WINDOW SIZE

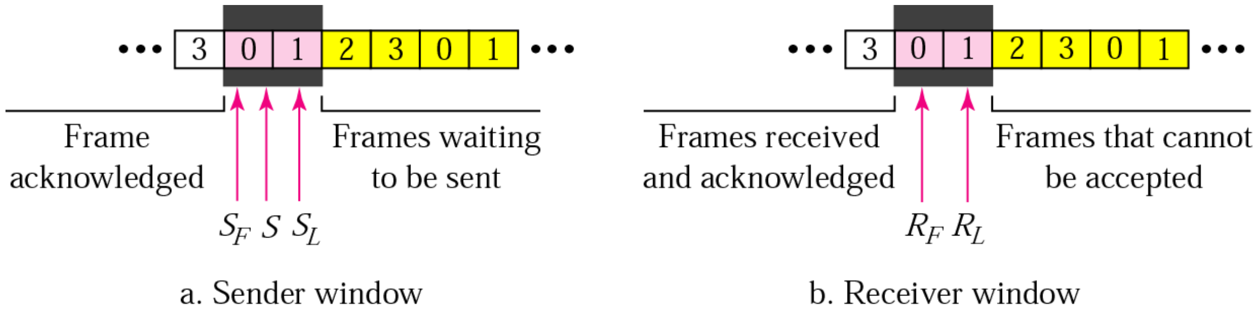
- Size of the sender window must be less than 2^m. Size of the receiver is always 1. If m = 2, window size = $2^m - 1 = 3$.
- Fig compares a window size of 3 and 4.



a. Window size < 2^m b. Window size = 2^m

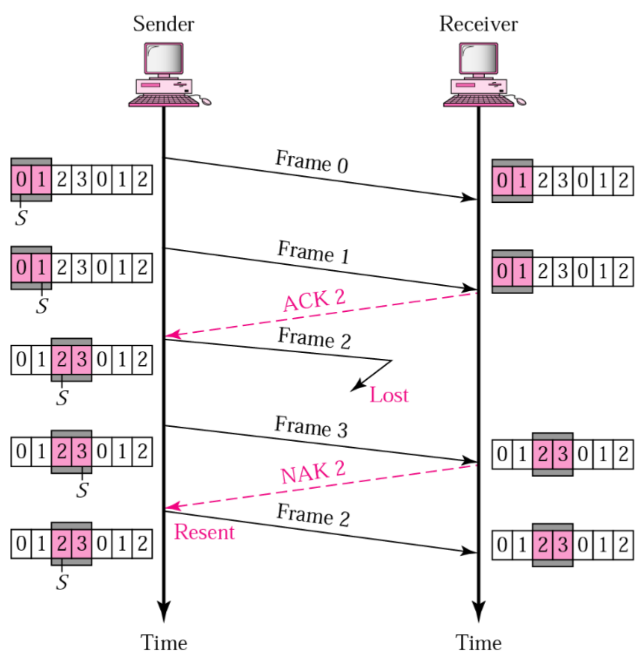
SELECTIVE REPEAT ARQ, SENDER AND RECEIVER WINDOWS

- Go-Back-N ARQ simplifies the process at the receiver site. Receiver only keeps track of only one variable, and there is no need to buffer out-of-order frames, they are simply discarded.
- However, Go-Back-N ARQ protocol is inefficient for noisy link. It bandwidth inefficient and slows down the transmission.
- In Selective Repeat ARQ, only the damaged frame is resent. More bandwidth efficient but more complex processing at receiver.
- It defines a negative ACK (NAK) to report the sequence number of a damaged frame before the timer expires.



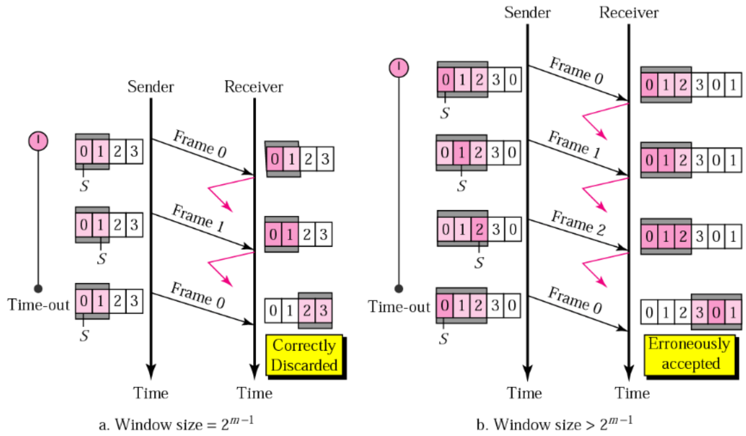
SELECTIVE REPEAT ARQ, LOST FRAME

- Frames 0 and 1 are accepted when received because they are in the range specified by the receiver window. Same for frame 3.
- Receiver sends a NAK2 to show that frame 2 has not been received and then sender resends only frame 2 and it is accepted as it is in the range of the window.



SELECTIVE REPEAT ARQ, SENDER WINDOW SIZE

- Size of the sender and receiver windows must be at most one-half of 2^m. If m = 2, window size should be $2^m/2 = 2$. Fig compares a window size of 2 with a window size of 3. Window size is 3 and all ACKs are lost, sender sends duplicate of frame 0, window of the receiver expect to receive frame 0 (part of the window), so accepts frame 0, as the 1st frame of the next cycle – an **error**.



3.3 Data link protocols: PPP, HDLC

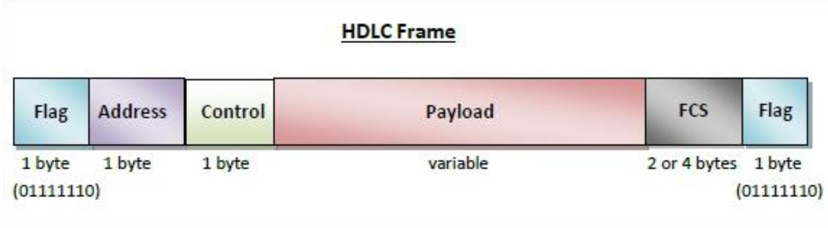
HIGH-LEVEL DATA LINK CONTROL (HDLC)

- High-level Data Link Control (HDLC) is a group of communication protocols of the data link layer for transmitting data between network points or nodes.
- The most important data link control protocol (ISO 3009, ISO 4335)
- Widely used
- Also, basis for many other important data link control protocols, which use the same or similar formats
- Uses synchronous transmission

TRANSFER MODES IN HDLC

- **Normal Response Mode (NRM)** − Here, two types of stations are there, a primary station that send commands and secondary station that can respond to received commands. It is used for both point - to - point and multipoint communications.
- **Asynchronous Balanced Mode (ABM)** − Here, the configuration is balanced, i.e. each station can both send commands and respond to commands. It is used for only point - to - point communications.

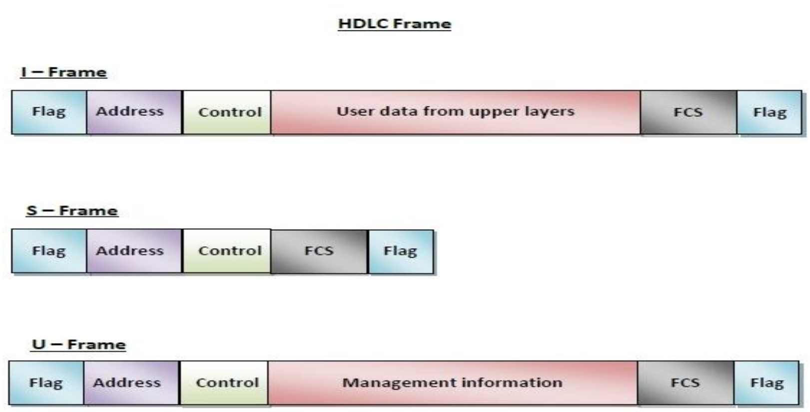
HDLC FRAME



- HDLC is a bit - oriented protocol where each frame contains up to six fields. The structure varies according to the type of frame. The fields of a HDLC frame are :
  - **Flag** − It is an 8-bit sequence that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
  - **Address** − It contains the address of the receiver. If the frame is sent by the primary station, it contains the address(es) of the secondary station(s). If it is sent by the secondary station, it contains the address of the primary station. The address field may be from 1 byte to several bytes.
  - **Control** − It is 1 or 2 bytes containing flow and error control information.
  - **Payload** − This carries the data from the network layer. Its length may vary from one network to another.
  - **FCS** − It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

TYPES OF HDLC FRAMES

- **I-frame** − I-frames or Information frames carry user data from the network layer. They also include flow and error control information that is piggybacked on user data. The first bit of control field of I-frame is 0.
- **S-frame** − S-frames or Supervisory frames do not contain information field. They are used for flow and error control when piggybacking is not required. The first two bits of control field of S-frame is 10.
- **U-frame** − U-frames or Un-numbered frames are used for myriad miscellaneous functions, like link management. It may contain an information field, if required. The first two bits of control field of U-frame is 11.



POINT-TO-POINT PROTOCOL (PPP)

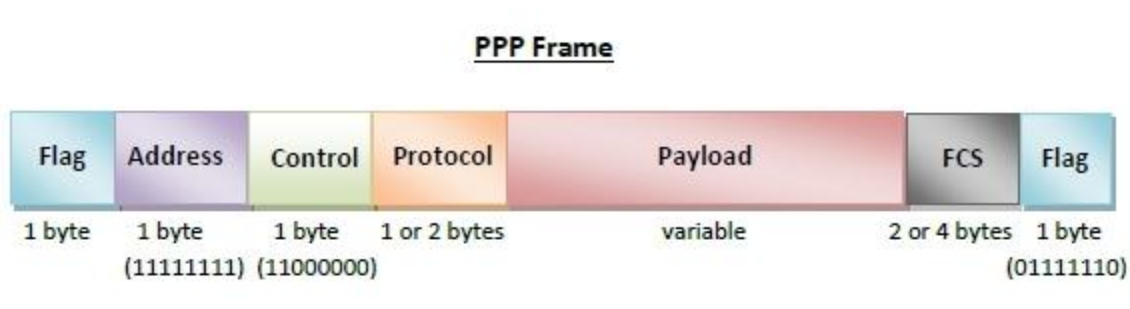
Point - to - Point Protocol (PPP) is a communication protocol of the data link layer that is used to transmit multiprotocol data between two directly connected (point-to-point) computers.

- It is a byte - oriented protocol that is widely used in broadband communications having heavy loads and high speeds.
- Since it is a data link layer protocol, data is transmitted in frames. It is also known as RFC 1661.

SERVICES PROVIDED BY PPP

- Defining the frame format of the data to be transmitted.
- Defining the procedure of establishing link between two points and exchange of data.
- Stating the method of encapsulation of network layer data in the frame.
- Stating authentication rules of the communicating devices.
- Providing address for network communication.
- Providing connections over multiple links.
- Supporting a variety of network layer protocols by providing a range OS services.

PPP FRAME



- **Flag** − 1 byte that marks the beginning and the end of the frame. The bit pattern of the flag is 01111110.
- **Address** − 1 byte which is set to 11111111 in case of broadcast.
- **Control** − 1 byte set to a constant value of 11000000.
- **Protocol** − 1 or 2 bytes that define the type of data contained in the payload field.
- **Payload** − This carries the data from the network layer. The maximum length of the payload field is 1500 bytes. However, this may be negotiated between the endpoints of communication.
- **FCS** − It is a 2 byte or 4 bytes frame check sequence for error detection. The standard code used is CRC (cyclic redundancy code)

COMPONENTS OF PPP

- **Encapsulation Component** − It encapsulates the datagram so that it can be transmitted over the specified physical layer.
- **Link Control Protocol (LCP)** − It is responsible for establishing, configuring, testing, maintaining and terminating links for transmission. It also imparts negotiation for set up of options and use of features by the two endpoints of the links.
- **Authentication Protocols (AP)** − These protocols authenticate endpoints for use of services. The two authentication protocols of PPP are:
    - Password Authentication Protocol (PAP)
    - Challenge Handshake Authentication Protocol (CHAP)

3.4 Logical link control (LLC) and media access control (MAC) sublayers

Logical link control (LLC)

- It adds header, trailer and acknowledgement number.
- It provides three services options:
    - Unacknowledged connectionless service (or Unreliable Datagram Services): It consists of having the source machine send independent frames to the destination machine without having the destination machine acknowledge them. This is appropriate when the error rate is very low. Example: Ethernet
    - Acknowledged connectionless service (or Acknowledged Datagram Service): It is useful for unreliable channels, such as wireless system. No logical connection is used, but each frame sent is individually acknowledged.
    - Acknowledged connection-oriented service (or Reliable Connection-oriented Services): The source and destination machines establish a connection before any data are transferred. Each frame over the connection is numbered, and the data link layer guarantees that each frame sent is indeed received.

Media access control (MAC)

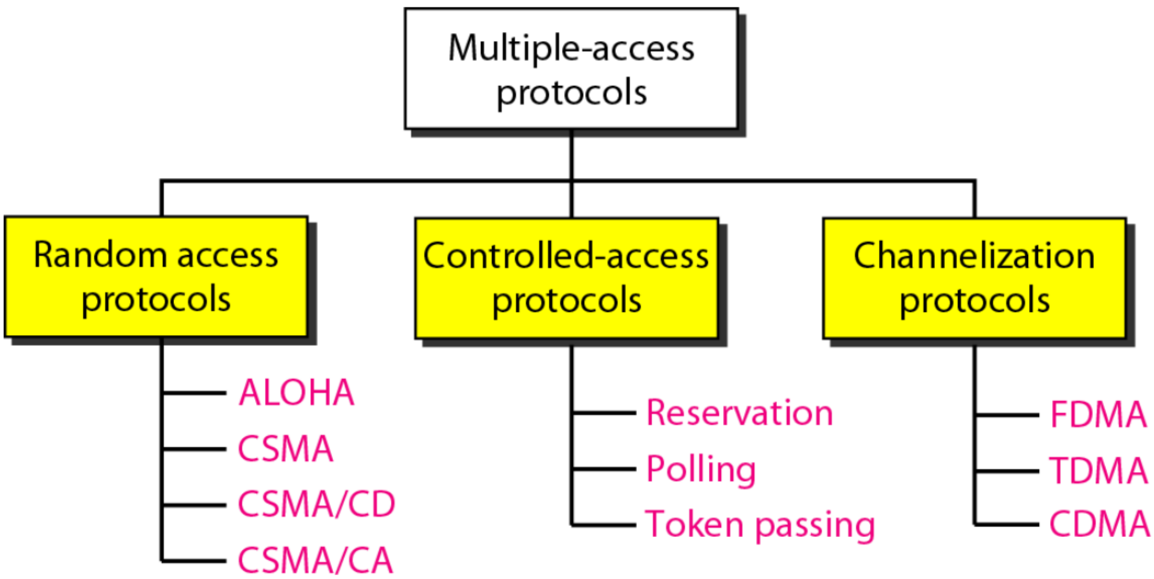
- The medium access control (MAC) is a sublayer of the data link layer of the open system interconnections (OSI) reference model for data transmission.
- It is responsible for flow control and multiplexing for transmission medium.
- It controls the transmission of data packets via remotely shared channels.
- It sends data over the network interface card.

Function of MAC sublayer

- It provides an abstraction of the physical layer to the LLC and upper layers of the OSI network.
- It is responsible for encapsulating frames so that they are suitable for transmission via the physical medium.
- It resolves the addressing of source station as well as the destination station, or groups of destination stations.

- It performs multiple access resolutions when more than one data frame is to be transmitted. It determines the channel access methods for transmission.
- It also performs collision resolution and initiating retransmission in case of collisions.
- It generates the frame check sequences and thus contributes to protection against transmission errors.

Multiple Access Control



Channelization

- Channelization is a multiple-access method in which the available bandwidth of a link is shared in time, frequency, or through code, between different stations
- Different channelization protocols:
- FDMA
- TDMA and
- CDMA

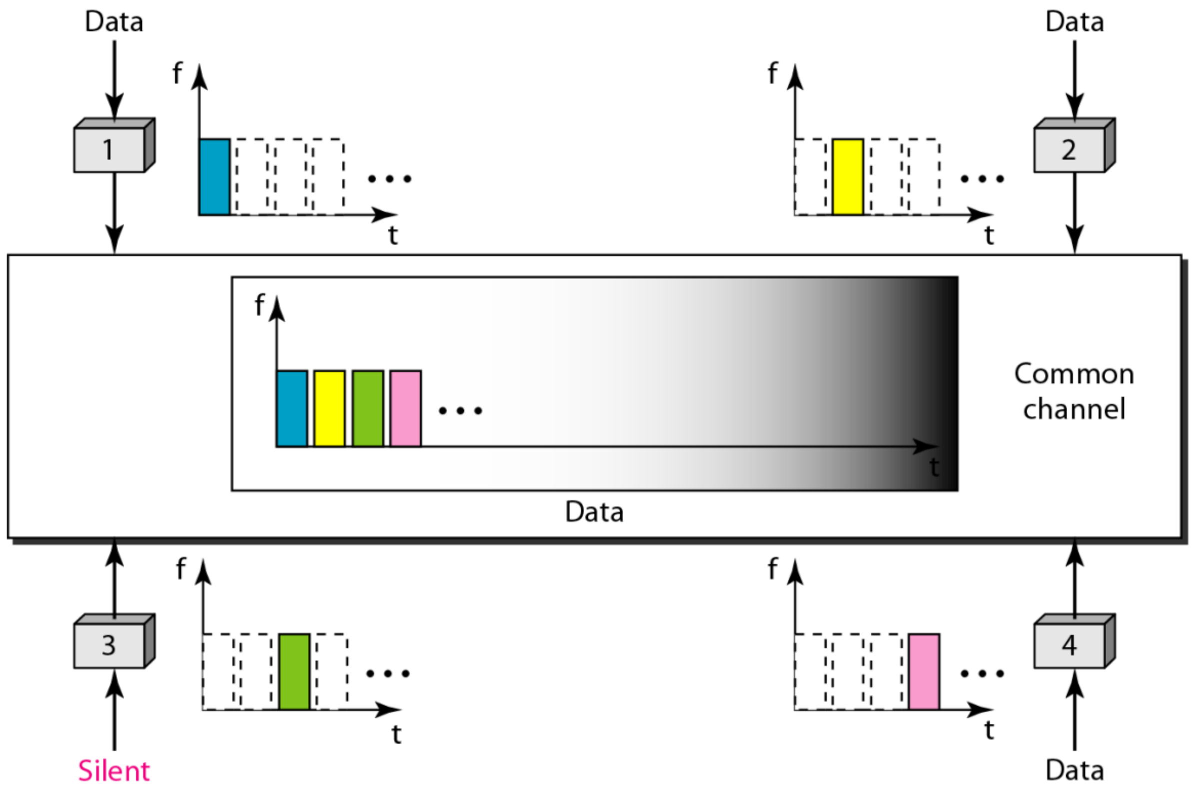
Frequency-division multiple access (FDMA)

- The available bandwidth is divided into frequency bands and each station is allocated a band to send its data
- Each band is reserved for a specific station, and it belongs to the station all the time
- To prevent station interferences, the allocated bands are separated from one another by small guard bands
- Specifies a predetermined frequency band for the entire period of communication

Time-division multiple access (TDMA)

- The stations share the bandwidth of the channel in time, but each station is allocated a time slot during which it can send data
- Each station transmits its data in is assigned time slot
- Need of synchronization between the different stations
- Each station needs to know the beginning of its slot and the location of its slot
- Synchronization is normally accomplished by having some synchronization bits (normally referred to as preamble bits) at the beginning of each slot

TDMA



Code-Division Multiple Access (CDMA)

- The concept of code-division multiple access (CDMA) was conceived several decades ago
- Recent advances in electronic technology have finally made its implementation possible
- CDMA differs from FDMA because only one channel occupies the entire bandwidth of the link
- It differs from TDMA because all stations can send data simultaneously; there is no timesharing
- In CDMA, one channel carries all transmissions simultaneously
- CDMA means communication with different codes

3.5 Random access protocols

- In this, all stations have same superiority i.e, no station has more priority than another station. Any station can send data depending on medium's state(idle or busy).
- It has two features:

  - There is no fixed time for sending data
  - There is no fixed sequence of stations sending data

Random Access:

- ALOHA
- Carrier Sense Multiple Access (CSMA)
- Carrier Sense Multiple Access with Collision Detection (CSMA/CD)
- Carrier Sense Multiple Access with Collision Avoidance (CSMA/CA)

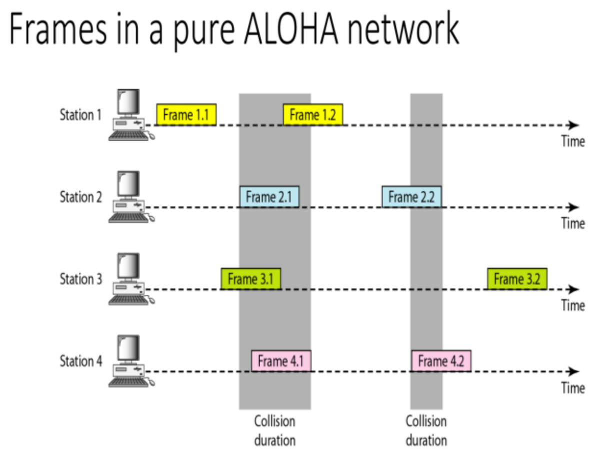
❖ ALOHA

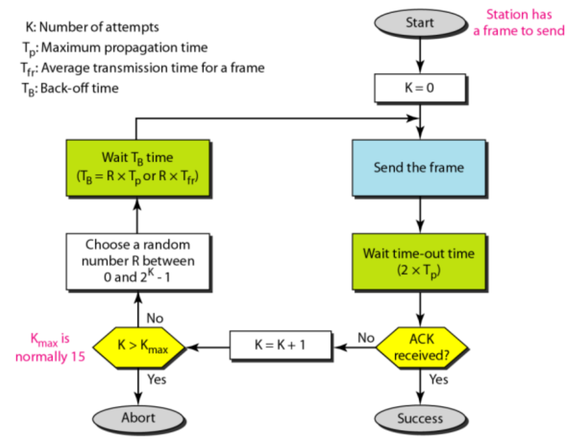
- The earliest random access method (first)
- Developed at the University of Hawaii in early 1970
- Designed for a radio (wireless) LAN, but can be used on any shared medium
- The medium is shared between stations
- The data from multiple stations can collide and become garbled
- Two versions of the protocol now called ALOHA:

  - Pure ALOHA and
  - Slotted ALOHA

❖ Pure ALOHA

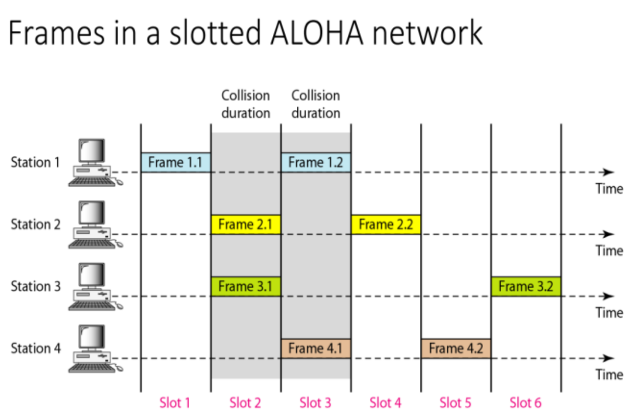
- In original ALOHA protocol each station is free to transmit
- Each station (source or transmitter) in a network sends data whenever there is a frame to send
- As there is only one channel, there might be collisions between frames from different stations
- Senders need some way to address such cases
- It relies on the acknowledgements from the receiver
- If the acknowledgement does not arrive after time-out period the station resends the frame

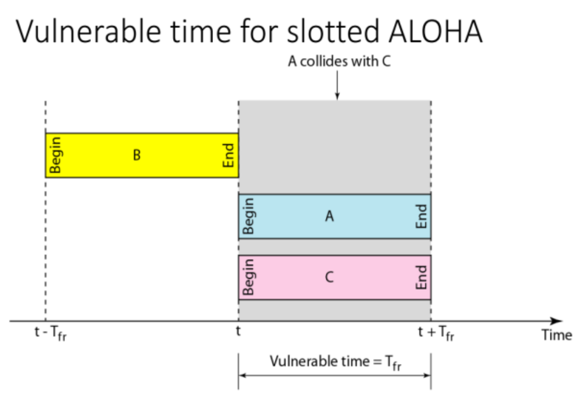




Slotted ALOHA

- In pure ALOHA there is no rule to define when a station can send
- A station can send soon after another station has started or soon before another station has finished
- Slotted ALOHA was invented to improve the efficiency of pure ALOHA
- Divide time into discrete intervals called slots
- Each interval corresponds to one frame T_{fr}
- All devices have to agree on slot boundaries i.e. any station can send only at the beginning of the time slot
- Synchronization is required:
- One special station emit a pip at the start of each interval, like clock



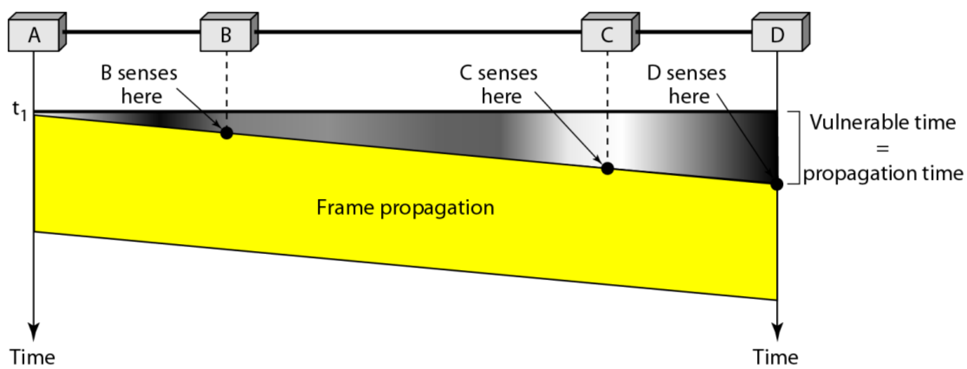


Carrier Sense Multiple Access (CSMA)

The performance can be improved by introducing carrier sensing (CS)

- With carrier sensing, each host listens to the data being transmitted over the channel before sending data by itself
- A host will only transmit its own frames when it cannot hear any data being transmitted by other hosts "Listen before talk"
- Decreases the chances of collisions when two or more stations start sending their signals via common channel
- The persistence methods can be applied to help the station take action when the channel is busy

Vulnerable time in CSMA



CSMA/CD

- Best Known Scheme for controlling a LAN is CSMA/CD
- Most Widely used implementation of CSMA/CD is found in Ethernet Specification.
- improvement is for stations to abort their transmissions as soon as they detect a collision.
- In other words, if two stations sense the channel to be idle and begin transmitting simultaneously, they will both detect the collision almost immediately. 64 CSMA/CD
- Rather than finish transmitting their frames, which are irretrievably garbled anyway, they should abruptly stop transmitting as soon as the collision is detected.
- Quickly terminating damaged frames saves time and bandwidth.
- This protocol, known as CSMA/CD (CSMA with Collision Detection) is widely used on LANs in the MAC sublayer.

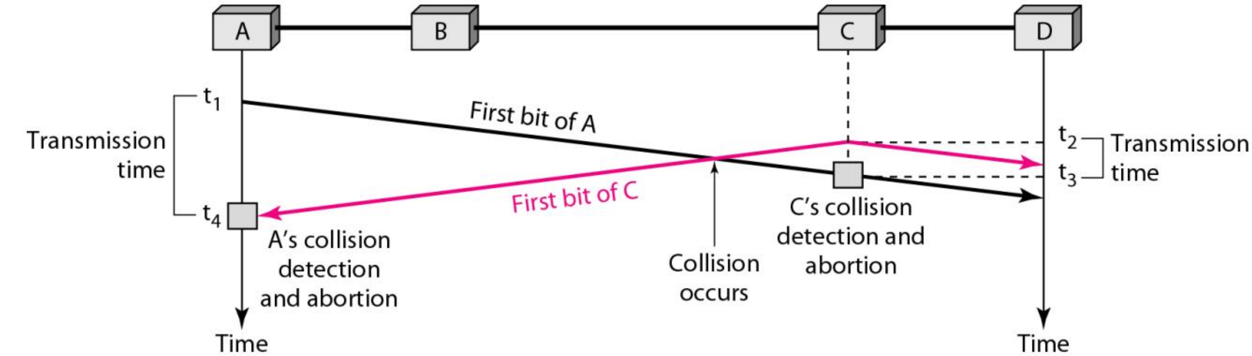


Fig: Collison of First bit in CSMA/CD

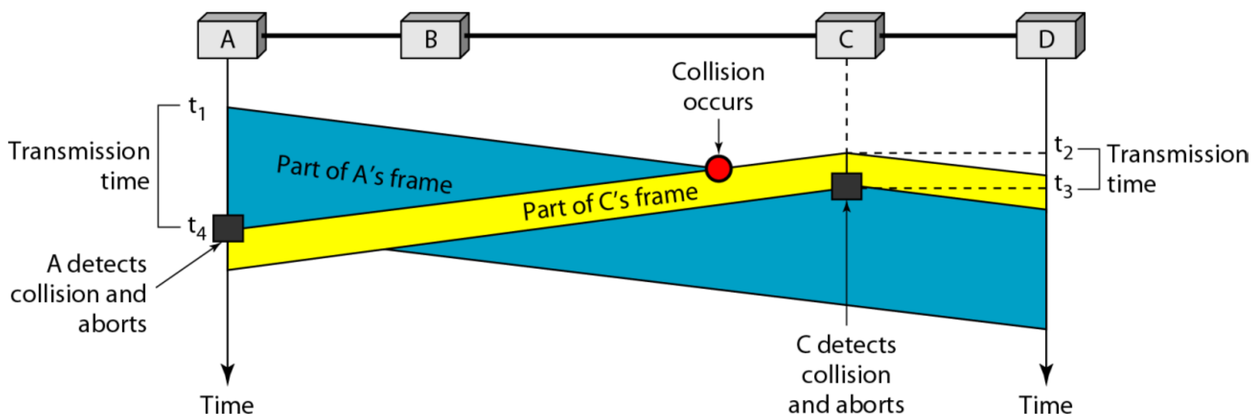


Fig: Collision and abortion in CSMA/CD

CSMA/CA:

- Carrier sense multiple access with collision avoidance.
- In wireless much of the sent energy is lost in transmission so the received signal has very little energy
- A collision may add only 5 to 10 percent of additional energy so not useful for detection of collision effectively
- As collisions cannot be detected it is necessary to avoid collisions on wireless networks
- CSMA/CA was invented for such type of networks
- The process of collisions detection involves sender receiving acknowledgement signals.
- If there is just one signal(its own) then the data is successfully sent but if there are two signals(its own and the one with which it has collided) then it means a collision has occurred.

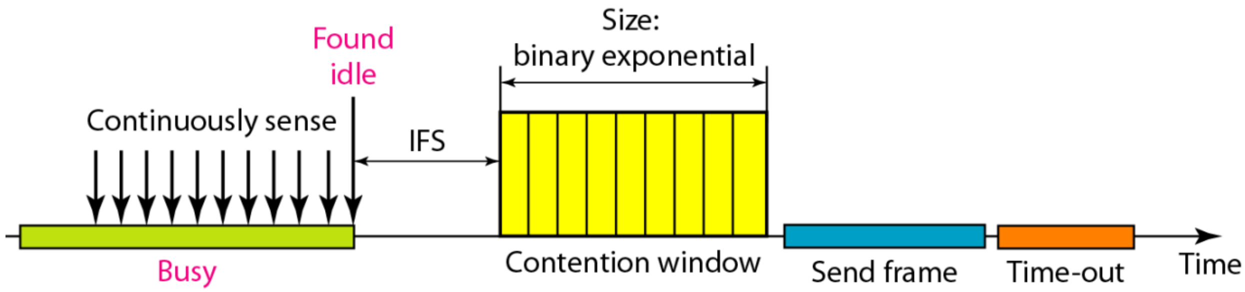


Fig: Timing in CSMA/CA

Controlled Access Protocol

- In this method, the stations consult each other to find which station has a right to send.
- A station cannot send unless it has beenauthorized by other station.
- The different controlled access methods are:
  - Reservation
  - Polling
  - Token Passing

Reservation

- In this method, a station needs to make a reservation before sending data.
- The time is divided into intervals. In eachinterval, a reservation frame precedes the data frames sent in that interval.
- When a station needs to send a frame, it makes a reservation in its own slot.
- The stations that have made reservations can send their frames after the reservation frame.

Polling

- Polling method works in those networks where primary and secondary stations exist.
- All data exchanges are made through primarydevice even when the final destination is a secondary device.
- Primary device controls the link and secondary device follow the instructions.

Token Passing

- Token passing method is used in those networks where the stations are organized in a logical ring.
- In such networks, a special packet called token is circulated through the ring.
- Whenever any station has some data to send, itwaits for the token. It transmits data only after it gets the possession of token.
- After transmitting the data, the station releases the token and passes it to the next station in the ring.

- If any station that receives the token has no data tosend, it simply passes the token to the next station in the ring.

SWITCHES

- A switch is a multiport bridge with a buffer and a design that can boost its efficiency(a large number of ports imply less traffic) and performance.
- A switch is a data link layer device.
- The switch can perform error checking before forwarding data, that makes it very efficient as it does not forward packets that have errors and forward good packets selectively to correct port only.

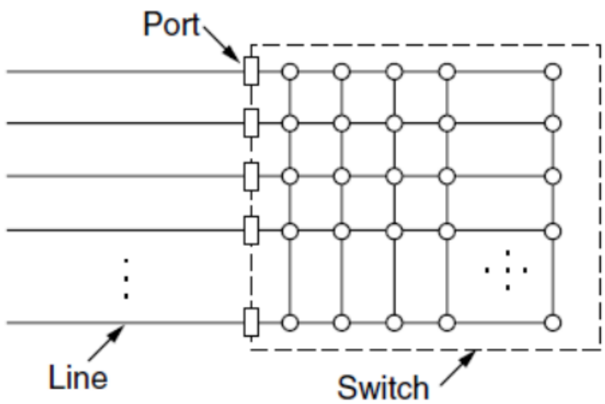


Fig: Switches

Bridges

- A bridge operates at data link layer.
- A bridge is a repeater, with add on the functionality of filtering content by reading the MAC addresses of source and destination.
- It is also used for interconnecting two LANs working on the same protocol.
- It has a single input and single output port, thus making it a 2 port device.

TYPES OF BRIDGES

- **Transparent Bridges: -** These are the bridge in which the stations are completely unaware of the bridge's existence i.e. whether or not a bridge is added or deleted from the network, reconfiguration of the stations is unnecessary. These bridges make use of two processes i.e. bridge forwarding and bridge learning.

- **Source Routing Bridges: -** In these bridges, routing operation is performed by source station and the frame specifies which route to follow. The host can discover frame by sending a special frame called discovery frame, which spreads through the entire network using all possible paths to destination.

3.6 Token-Based Protocols

Token-based protocols use a special control frame called a token. Only the station holding the token can transmit data. This avoids collisions.

1. Token Bus

- Defined under **IEEE 802.4**
- Logical ring formed over a physical bus topology.
- A token circulates in a predefined order.
- A station transmits only when it receives the token.
- Used in industrial environments.

Advantage: No collision
Disadvantage: Complex management

2. Token Ring

- Defined under **IEEE 802.5**
- Physical ring topology.
- Token passes sequentially from one station to another.
- Only token holder transmits.

Advantages: Predictable performance, no collision
Disadvantages: If ring breaks, network may stop

3. FDDI (Fiber Distributed Data Interface)

- Uses optical fiber.
- Dual ring structure (primary + secondary for backup).
- High speed (100 Mbps).
- Suitable for backbone networks.

Advantage: Fault tolerance
Disadvantage: Expensive

3.7 Ethernet (IEEE 802.3) and Its Evolution

Ethernet (IEEE 802.3)

- Defined under **IEEE 802.3**
- Uses CSMA/CD in traditional LANs.
- Originally bus topology, now star topology with switches.
- Frame size: 64–1518 bytes.

Evolution of Ethernet

- 10 Mbps (Standard Ethernet)
- 100 Mbps (Fast Ethernet)
- 1 Gbps (Gigabit Ethernet)
- 10 Gbps and above (Modern Ethernet)

Improved speed, reliability, and switching technology over time.

Bridged Ethernet

- Uses **bridges** to connect multiple LAN segments.
- Reduces collision domains.
- Operates at Data Link Layer.

Switched Ethernet

- Uses **switches** instead of hubs.
- Each port has separate collision domain.
- Supports full-duplex communication.
- Much faster and efficient than bridged Ethernet.

3.8 WLAN (Wi-Fi) and IEEE 802.11

WLAN (Wireless LAN)

- Provides wireless network connectivity.
- Uses radio waves.
- Commonly called Wi-Fi.

Defined under **IEEE 802.11**

Features:

- Uses CSMA/CA (collision avoidance).
- Operates in 2.4 GHz and 5 GHz bands.
- Supports mobility.

Evolution:

- 802.11b (11 Mbps)
- 802.11g (54 Mbps)
- 802.11n (Higher speed, MIMO)
- 802.11ac / ax (Very high speed)

3.9 VLAN (Virtual Local Area Network)

A VLAN is a **logical grouping of devices** within a network, regardless of physical location.

Why VLAN is Important:

- Improves security
- Reduces broadcast traffic
- Better network management
- Separates departments logically (e.g., HR, Accounts)

Works at Data Link Layer using VLAN tagging (IEEE 802.1Q).

3.10 Virtual Circuit Switching

Virtual circuit switching establishes a logical path before data transfer begins.

Two important technologies:

1. **ATM (Asynchronous Transfer Mode)**
    - Uses fixed-size cells (53 bytes).
    - High speed switching.
    - Connection-oriented.
    - Used in backbone networks.

2. MPLS (Multiprotocol Label Switching)

- Adds label to packets for faster forwarding.
- Works between Layer 2 and Layer 3.
- Used in modern ISP networks.
- Improves traffic engineering and QoS.